

LANSING, MI, USA

WMO: 725390

Lat: 42.776N

Lon: 84.600W

Elev: 262

StdP: 98.22

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14836

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-15.4	-22.0	0.5	-17.5	-19.2	0.7	-14.6	11.9	-3.3	11.0	-3.3	2.8	280	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	31.8	22.7	30.2	21.9	28.8	21.0	24.4	29.4	23.4	28.1	22.4	26.9	5.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.0	27.3	21.8	17.0	26.1	20.9	16.1	25.2	74.7	29.5	70.7	28.2	67.0	26.8	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.1	8.3	-22.7	34.3	3.6	1.7	-25.3	35.5	-27.4	36.4	-29.4	37.4	-32.0	38.6		
(5)			-22.9	26.0	3.5	1.3	-25.4	26.9	-27.5	27.7	-29.4	28.4	-32.0	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-4.9	-3.7	1.5	8.2	14.4	19.9	21.9	20.9	17.0	10.4	4.0	-1.6		
	DBStd	10.67	6.36	6.10	6.28	5.15	4.83	3.88	3.30	3.20	4.30	4.85	5.17	5.40		
	HDD10.0	1833	462	383	274	94	12	0	0	0	2	54	191	360		
	HDD18.3	3723	719	616	522	306	141	28	7	12	75	249	430	617		
	CDD10.0	1497	1	1	11	39	149	296	368	338	213	68	11	1		
	CDD18.3	346	0	0	1	2	20	73	117	93	36	4	0	0		
	CDH23.3	3111	0	0	6	25	204	682	1100	747	311	36	0	0		
	CDH26.7	904	0	0	1	3	47	212	361	196	80	4	0	0		
(14) Wind	WSAvg	3.9	4.6	4.5	4.3	4.5	3.9	3.4	3.1	2.9	3.1	3.8	4.2	4.3		
(15) Precipitation	PrecAvg	776	39	36	56	74	75	89	67	79	84	55	65	53		
	PrecMax	1005	85	97	111	147	265	259	163	249	212	144	143	108		
	PrecMin	539	6	6	4	17	16	5	13	4	0	11	13	9		
	PrecStd	108	22	23	25	30	49	48	30	41	49	31	31	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.5	23.4	26.5	30.2	33.0	34.1	33.0	31.5	26.9	20.1	15.0		
		MCWB	11.6	11.2	16.6	16.4	20.5	23.5	24.2	23.3	22.0	19.8	14.1	13.0		
	2%	DB	9.4	10.8	18.4	22.9	27.9	31.0	32.1	30.5	29.0	23.8	16.6	11.1		
		MCWB	7.7	7.8	13.5	14.9	19.2	21.9	23.4	22.5	20.9	17.2	12.5	9.3		
	5%	DB	6.0	7.0	14.5	20.2	25.7	29.3	30.3	28.8	26.9	21.3	14.5	8.3		
		MCWB	4.4	4.6	10.4	13.0	18.1	21.0	22.3	21.4	19.7	16.3	11.0	6.4		
	10%	DB	3.3	4.4	11.1	17.5	23.3	27.3	28.6	27.3	24.8	18.9	12.2	5.4		
		MCWB	2.1	2.4	7.4	11.9	16.9	20.0	21.2	20.7	18.6	14.6	9.1	3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	12.3	16.8	19.0	22.3	24.6	26.3	25.3	23.9	21.2	15.6	13.4		
		MCDB	12.9	13.6	22.9	24.7	28.2	30.9	31.6	30.7	29.4	25.7	18.4	15.1		
	2%	WB	8.0	8.1	13.9	16.1	20.6	23.2	24.8	23.9	22.3	19.0	13.6	9.8		
		MCDB	9.0	10.3	17.5	20.4	25.5	28.8	30.0	28.1	26.2	22.2	15.8	11.2		
	5%	WB	4.5	4.9	11.2	14.2	19.4	22.2	23.7	22.8	21.1	16.9	11.4	6.4		
		MCDB	5.8	6.8	14.2	19.1	24.0	27.4	28.5	26.9	24.8	20.3	14.0	7.9		
	10%	WB	2.1	2.5	7.7	12.1	17.8	21.2	22.6	21.8	19.8	15.1	9.1	3.7		
		MCDB	3.3	4.3	10.8	16.8	22.2	25.6	27.0	25.8	23.8	18.5	11.9	5.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.2	9.8	11.5	11.5	11.3	11.7	11.0	11.7	10.2	8.1	6.5		
		MCDBR	9.1	10.3	13.8	15.0	13.5	12.9	12.6	12.0	13.4	13.4	11.1	9.4		
	5% WB	MCWBR	8.0	8.5	9.6	8.9	7.1	6.0	6.0	5.5	6.7	8.1	8.1	8.1		
		MCWBR	8.7	9.7	12.8	13.2	11.7	11.0	11.1	10.0	11.0	11.7	10.1	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.350	0.337	0.384	0.414	0.436	0.425	0.419	0.381	0.344	0.314	0.294		
	taud		2.359	2.163	2.370	2.261	2.227	2.202	2.239	2.250	2.345	2.445	2.476	2.414		
	Ebn at Noon		841	830	905	882	861	840	845	838	847	840	818	810		
	Edh at Noon		81	116	107	130	138	142	135	130	110	88	72	70		
(44) All-Sky Solar Radiation	RadAvg		1.43	2.32	3.59	4.57	5.34	6.03	6.12	5.24	4.21	2.68	1.66	1.14		
	RadStd		0.13	0.29	0.37	0.42	0.50	0.37	0.29	0.25	0.40	0.27	0.22	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+133	+60
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page